



DICOM Conformance Statement

DICOM Forwarder

Version 1.5

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1 Conformance Statement Overview

This document is structured as suggested in the DICOM Standard (PS 3.2: Conformance).

Table 1-1 Network Services Supported

SOP Classes	User of Service (SCU)	Provider of Service (SCP)
Verification		
Verification	Yes	Yes
Transfer Image Storage SOP Classes		
Secondary Capture Image Storage	Yes	Yes
Multi-frame Grayscale Byte Secondary Capture Image Storage	Yes	Yes
Multi-frame True Color Secondary Capture Image Storage	Yes	Yes
Visible Light Endoscopic Image Storage	Yes	Yes
Visible Light Microscopic Image Storage	Yes	Yes
Visible Light Photographic Image Storage	Yes	Yes
Ophthalmic Photography 8 Bit Image Storage	Yes	Yes
Ophthalmic Tomography Image Storage	Yes	Yes
Transfer Video Image Storage SOP Classes		
Ophthalmic Photography 8 Bit Image Storage	Yes	Yes
Video Endoscopic Image Storage	Yes	Yes
Video Microscopic Image Storage	Yes	Yes
Video Photographic Image Storage	Yes	Yes
Transfer Other Storage SOP Classes		
Encapsulated PDF Storage	Yes	Yes
Raw Data Storage	Yes	Yes
Lensometry Measurements Storage	Yes	Yes
Autorefractive Measurements Storage	Yes	Yes
Keratometry Measurements Storage	Yes	Yes
Subjective Refraction Measurements Storage	Yes	Yes
Visual Acuity Measurements Storage	Yes	Yes
Ophthalmic Axial Measurements Storage	Yes	Yes
Intraocular Lens Calculations Storage	Yes	Yes
Ophthalmic Visual Field Static Perimetry Measurements Storage	Yes	Yes
Grayscale Softcopy Presentation State Storage	Yes	Yes
Corneal Topography Map Storage	Yes	Yes
Enhanced SR Storage	Yes	Yes
Comprehensive SR Storage	Yes	Yes
Key Object Selection Document Storage	Yes	Yes
Ophthalmic Photography 16 Bit Image Storage	Yes	Yes
Spectacle Prescription Report Storage	Yes	Yes
Macular Grid Thickness and Volume Report	Yes	Yes

Ophthalmic Thickness Map Storage	Yes	Yes
Wide Field Ophthalmic Photography Stereographic Projection Image Storage	Yes	Yes
Wide Field Ophthalmic Photography 3D Coordinates Image Storage	Yes	Yes
Ophthalmic Optical Coherence Tomography En Face Image Storage	Yes	Yes
Ophthalmic Optical Coherence Tomography B-scan Volume Analysis Storage	Yes	Yes
Surface Segmentation Storage	Yes	Yes
Workflow Management		
Storage Commitment Push Model SOP Class	No	Yes
Modality Worklist Information Model – FIND	No	Yes
Query / Retrieve		
Patient Root Query/Retrieve Information Model – FIND	No	Yes
Study Root Query/Retrieve Information Model – FIND	No	Yes
Study Root Query/Retrieve Information Model – MOVE	No	Yes

DICOM Forwarder does not support Media Interchange.

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3 Introduction

3.1 Revision History

Table 3-1 Revision History

Document Version	Date	Changes
I	2021-04-26	Creation of document for DICOM Forwarder v1.0
1.1	2021-10-18	Added QIDO—RS, WADO-RS, STOW-RS status codes, added the DF specific implementation class UID
1.2	2022-11-03	Extended list of attributes and supported status codes for QIDO—RS, WADO-RS, STOW-RS. Updated query, retrieve and storage activities.
1.3	2023-05-03	Extended list of supported SOP Class UIDs. Documented TLS Support in Security section. Added description for lenient mode in case of Retrieve. Created configuration parameters table with DICOM connection settings under configuration section. Adapted note for query match key Modalities In Study. Extended list of MWL query return keys with new attributes. Added table for supported private query attributes. Added http error code mappings for 403, 500 and 404 to Store, Retrieve and MWL status code tables.
1.4	2023-10-18	Added tables with the list of required and optional attributes considered during store validation under Activity – DICOM Forwarder Issues Storage Request.
1.5	2024-05-15	Added new remarks under Table 4 36 MWL and UPS query match key mappings due to enhancements in MWL to UPS mapping. Added new Table 4 37 UPS to MWL query return key mappings with remarks due to enhancements in UPS to MWL mapping. Updated Accepted Character Sets as well as Returned Character Sets under Section 6.

3.2 Audience

This document is written for the people that need to understand how DICOM Forwarder will integrate into their healthcare facility. This includes both those responsible for overall imaging network policy and architecture, as well as integrators who need to have a detailed understanding of the DICOM features of the product. This document contains some basic DICOM definitions so that any reader may understand how this product implements DICOM features. However, integrators are expected to fully understand all the DICOM terminology, how the tables in this document relate to the product's functionality, and how that functionality integrates with other devices that support compatible DICOM features.

3.3 Remarks

The scope of this DICOM Conformance Statement is to facilitate integration between DICOM Forwarder and other DICOM products. The Conformance Statement should be read and understood in conjunction with the DICOM Standard. DICOM by itself does not guarantee interoperability. The Conformance Statement does, however, facilitate a first-level comparison for interoperability between different applications supporting compatible DICOM functionality.

This Conformance Statement is not supposed to replace validation with other DICOM equipment to ensure proper exchange of intended information. In fact, the user should be aware of the following important issues:

- The comparison of different Conformance Statements is just the first step towards assessing interconnectivity and interoperability between the product and other DICOM conformant equipment.
- Test procedures should be defined and executed to validate the required level of interoperability with specific compatible DICOM equipment, as established by the healthcare facility.

3.4 Definitions and Terms

Informal definitions are provided for the following terms used in this Conformance Statement.

The DICOM Standard is the authoritative source for formal definitions of these terms.

Abstract Syntax

The information agreed to be exchanged between applications, generally equivalent to a Service/Object Pair (SOP) Class.

Examples: Verification SOP Class, Modality Worklist Information Model Find SOP Class, Computed Radiography Image Storage SOP Class.

Application Entity (AE)

An end point of a DICOM information exchange, including the DICOM network or media interface software; i.e., the software that sends or receives DICOM information objects or messages. A single device may have multiple Application Entities.

Application Entity Title

The externally known name of an Application Entity, used to identify a DICOM application to other DICOM applications on the network.

Application Context

The specification of the type of communication used between Application Entities.

Example: DICOM network protocol.

Association

A network communication channel set up between Application Entities.

Attribute

A unit of information in an object definition; a data element identified by a tag. The information may be a complex data structure (Sequence), itself composed of lower-level data elements.

Examples: Patient ID (0010,0020), Accession Number (0008,0050), Photometric Interpretation (0028,0004), Procedure Code Sequence (0008,1032).

Information Object Definition (IOD)

The specified set of Attributes that comprise a type of data object; does not represent a specific instance of the data object, but rather a class of similar data objects that have the same properties. The Attributes may be specified as Mandatory (Type 1), Required but possibly unknown (Type 2), or Optional (Type 3), and there may be conditions associated with the use of an Attribute (Types 1C and 2C).

Examples: MR Image IOD, CT Image IOD, Print Job IOD.

Joint Photographic Experts Group (JPEG)

A set of standardized image compression techniques, available for use by DICOM applications.

Media Application Profile

The specification of DICOM information objects and encoding exchanged on removable media (e.g., CDs)

Module

A set of Attributes within an Information Object Definition that are logically related to each other.

Example: Patient Module includes Patient Name, Patient ID, Patient Birth Date, and Patient Sex.

Negotiation

First phase of Association establishment that allows Application Entities to agree on the types of data to be exchanged and how that data will be encoded.

Presentation Context

The set of DICOM network services used over an Association, as negotiated between Application Entities; includes Abstract Syntaxes and Transfer Syntaxes.

Protocol Data Unit (PDU)

A packet (piece) of a DICOM message sent across the network. Devices must specify the maximum size packet they can receive for DICOM messages.

Query Key

An input value for a query process. Query Keys denote the set of DICOM tags that are sent from the SCU to SCP and thus control the query result.

Security Profile

A set of mechanisms, such as encryption, user authentication, or digital signatures, used by an Application Entity to ensure confidentiality, integrity, and/or availability of exchanged DICOM data

Service Class Provider (SCP)

Role of an Application Entity that provides a DICOM network service; typically, a server that performs operations requested by another Application Entity (Service Class User).

Examples: Picture Archiving and Communication System (image storage SCP, and image query/retrieve SCP), Radiology Information System (modality worklist SCP).

Service Class User (SCU)

Role of an Application Entity that uses a DICOM network service; typically, a client.

Examples: imaging modality (image storage SCU, and modality worklist SCU), imaging workstation (image query/retrieve SCU)

Service/Object Pair (SOP) Class

The specification of the network or media transfer (service) of a particular type of data (object); the fundamental unit of DICOM interoperability specification.

Examples: Ultrasound Image Storage Service, Basic Grayscale Print Management.

Service/Object Pair (SOP) Instance

An information object; a specific occurrence of information exchanged in a SOP Class.

Examples: a specific x-ray image.

Tag

A 32-bit identifier for a data element, represented as a pair of four-digit hexadecimal numbers, the "group" and the "element". If the "group" number is odd, the tag is for a private (manufacturer-specific) data element.

Examples: (0010,0020) [Patient ID], (07FE,0010) [Pixel Data], (0019,0210) [private data element]

Transfer Syntax

The encoding used for exchange of DICOM information objects and messages.

Examples: JPEG compressed (images), little endian explicit value representation.

Unique Identifier (UID)

A globally unique "dotted decimal" string that identifies a specific object or a class of objects; an ISO-8824 Object Identifier.

Examples: Study Instance UID, SOP Class UID, SOP Instance UID.

Value Representation (VR)

The format type of an individual DICOM data element, such as text, an integer, a person's name, or a code. DICOM information objects can be transmitted with either explicit identification of the type of each data element (Explicit VR), or without explicit identification (Implicit VR); with Implicit VR, the receiving application must use a DICOM data dictionary to look up the format of each data element.

3.5 Abbreviations

Table 3-2 Abbreviations used in this Document

Abbreviation	Definition
AE	Application Entity
AET	Application Entity Title
DF	DICOM Forwarder
DICOM	Digital Imaging and Communications in Medicine
EBE	Explicit Big Endian
ELE	Explicit Little Endian
EMR	Electronic Medical Record
EPDF	Encapsulated Portable Document Format
HDP	Health Data Platform
HL7	Health Level Seven
ILE	Implicit Little Endian
IOD	Information Object Definition
MWL	Modality Worklist
MPEG2	Motion Picture Expert Group 2; Abbreviation and synonym for video encoding and compression transfer syntax.
MPEG4	Motion Picture Expert Group 4; Abbreviation and synonym for video encoding and compression transfer syntax.
MPML	MPEG2 Main Profile @ Main Level (as defined in the MPEG-2 standard)
MP HL	MPEG2 Main Profile @ High Level (as defined in the MPEG-2 standard)
RLE	Run Length Encoding
SCP	Service Class Provider
SCU	Service Class User
SOP	Service Object Pair, union of a specific DICOM service and related IOD.
TCP/IP	Transmission Control Protocol / Internet Protocol
UI	User Interface
UID	Unique Identifier
UPS	Unified Procedure Step
VL	Visible Light
XML	Extensible Markup Language

3.6 References

NEMA PS3 / ISO 12052, Digital Imaging and Communications in Medicine (DICOM) Standard, National Electrical Manufacturers Association, Rosslyn, VA, USA (available free at <http://medical.nema.org/>).

Integrating the Healthcare Enterprise (IHE) EYECARE Technical Framework, rev 3.7, 2010 (available free at http://www.ihe.net/Technical_Framework/index.cfm)

4 Networking

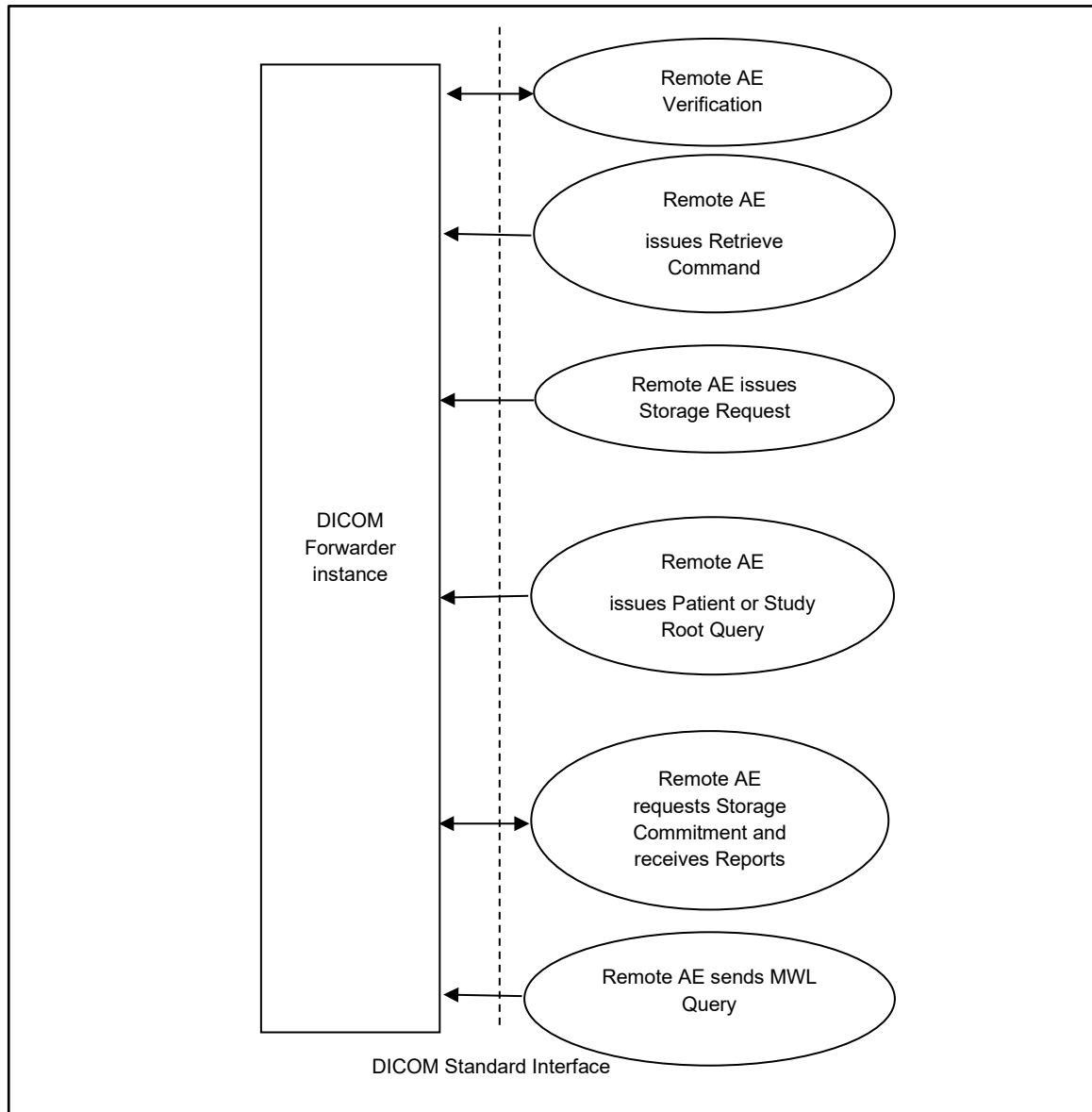
4.1 Implementation Model

The DICOM Forwarder “translates” DICOM DIMSE requests from a device to DICOMweb and forwards the request to a DICOMweb Server in the cloud.

As not all DICOM DIMSE functionalities are supported by DICOMweb standard, some get translated seamlessly to FHIR requests instead and are forwarded to a FHIR server.

4.1.1 Application Data Flow

Figure 4-1 DICOM Forwarder - Functional Overview



4.1.2 Functional Definition of AEs

4.1.2.1 Functional Definition of DICOM Forwarder Application Entity

DICOM Forwarder is a software for transferring DICOM data, e.g.: images, videos, and reports provided by computer-controlled diagnostic devices or other documentation systems.

The DICOM Forwarder AE initiates an association to a remote AE.

The DICOM Forwarder AE waits in the background for connections, and

- Answers Verification requests
- Processes Retrieve command and transfers data towards Remote AE.
- Processes and transforms Study and Patient Root Queries to HDP Cloud specific request.
- Processes Storage requests from remote AE and forwards them to HDP Cloud
- Processes and transfers Storage Commitment requests to HDP Cloud.
- Processes Storage Commitment requests, and therefore initiates an association and sends N-EVENT-REPORT-RQ.

4.1.3 Sequencing of Real-World Activities

To realize the real-world activities, the different entities work together. The sequence diagrams depict the intended workflow.

The only sequencing constraint that exists across DICOM Forwarder and the corresponding Application Entities, is the fact that a Composite SOP Instance must be received by DICOM Forwarder before Storage Commitment Push Model or Query-Retrieve Requests related to this SOP Instance can be successfully handled.

All Verification, Storage, Storage Commit, Modality Worklist Query and Retrieve DIMSE requests, as well as Study and Patient Root Query requests on Study, Series or Image level are converted to DICOMweb requests by the DICOM Forwarder and are forwarded to a DICOMweb Server hosted by HDP in the cloud.

However, as the DICOMweb standard does not support Patient Root Queries on Patient level, such DIMSE requests are translated seamlessly by DICOM Forwarder to FHIR requests instead and are forwarded to a FHIR server also hosted by HDP in the cloud as shown in **Figure 4-5**.

4.1.3.1 DICOM Forwarder Activities

Figure 4-2 Verification

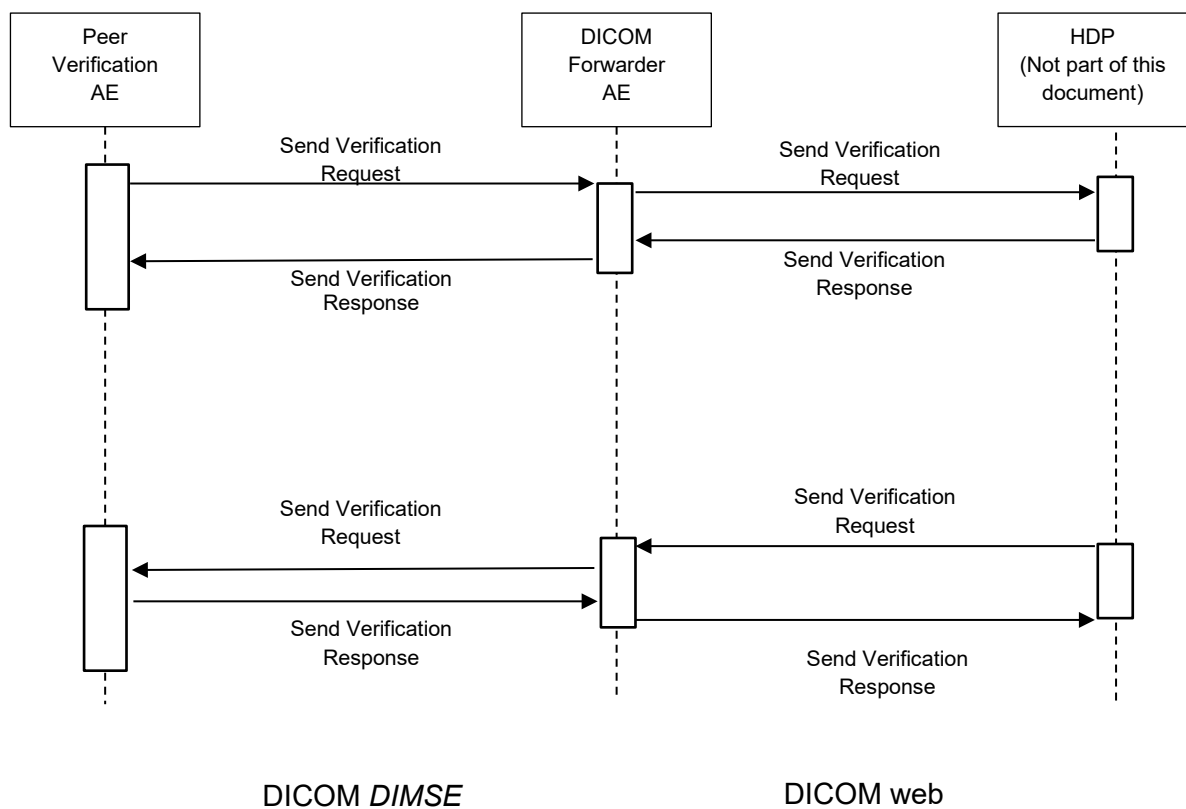
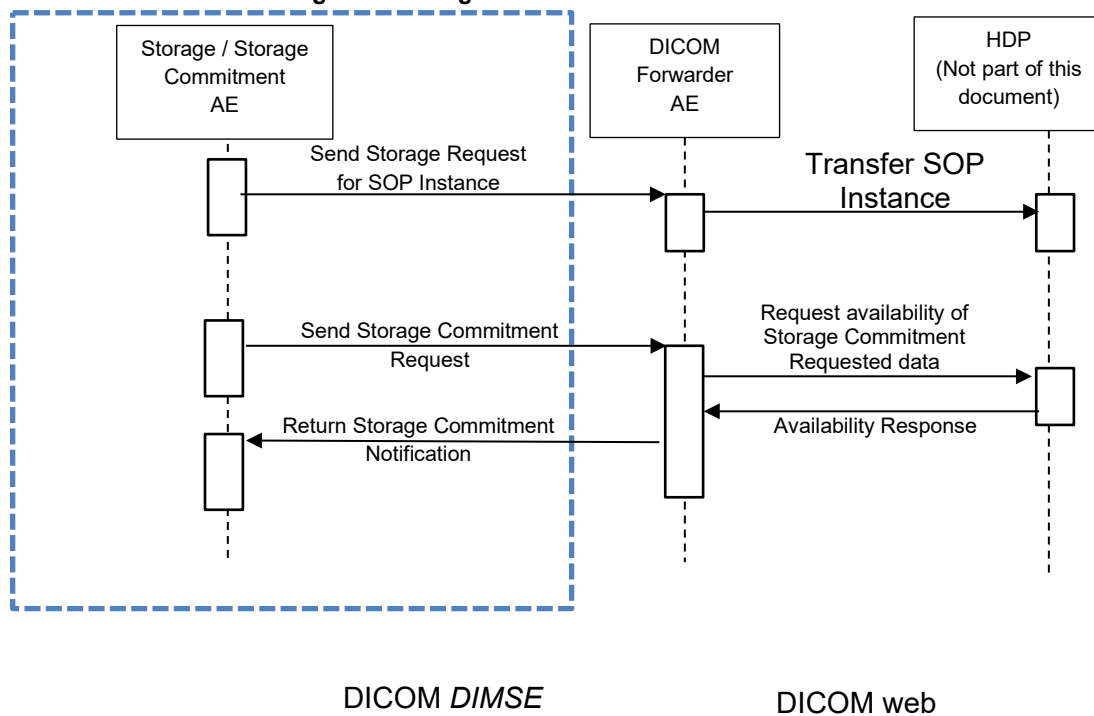
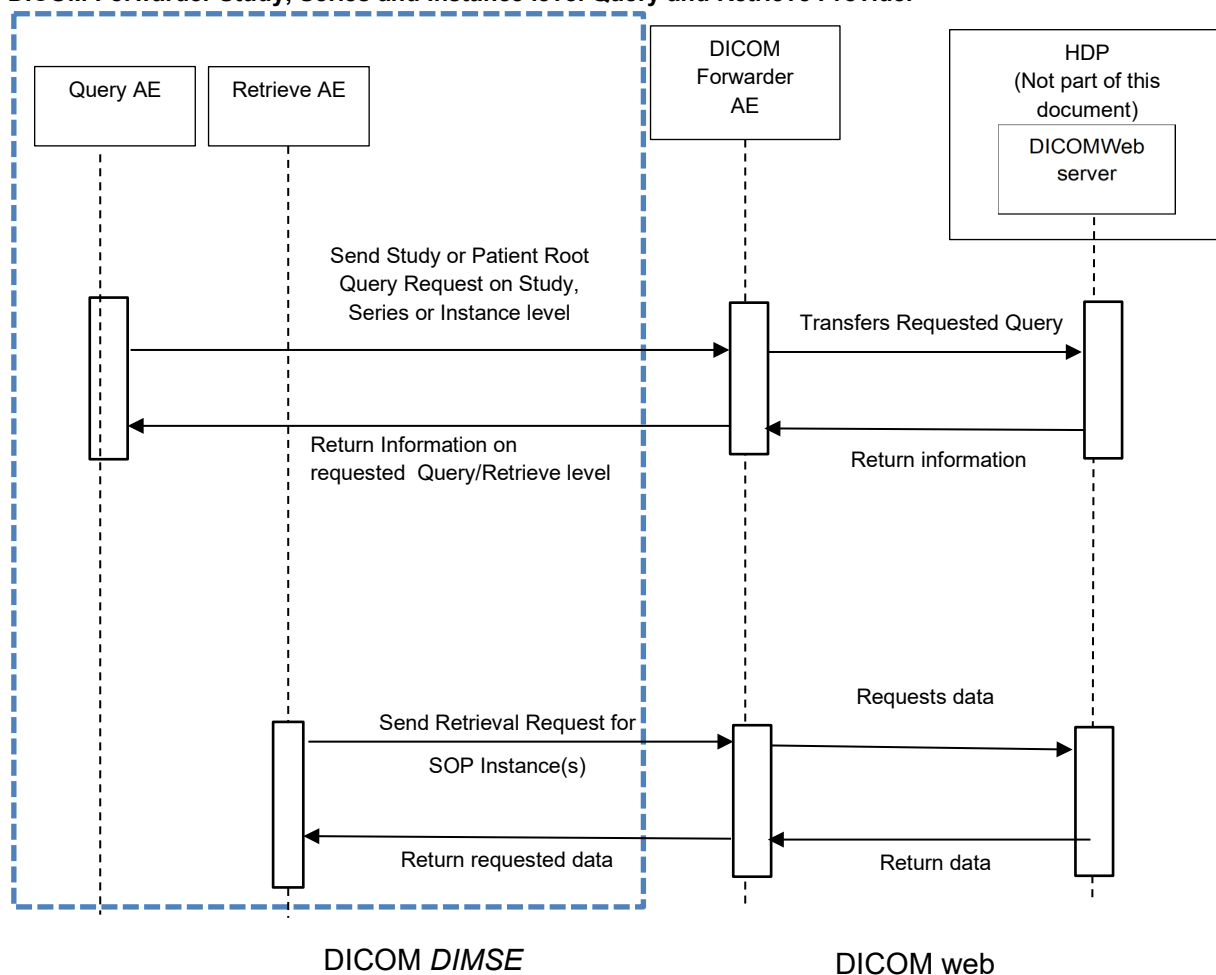


Figure 4-3 DICOM Forwarder Storage and Storage Commitment Provider



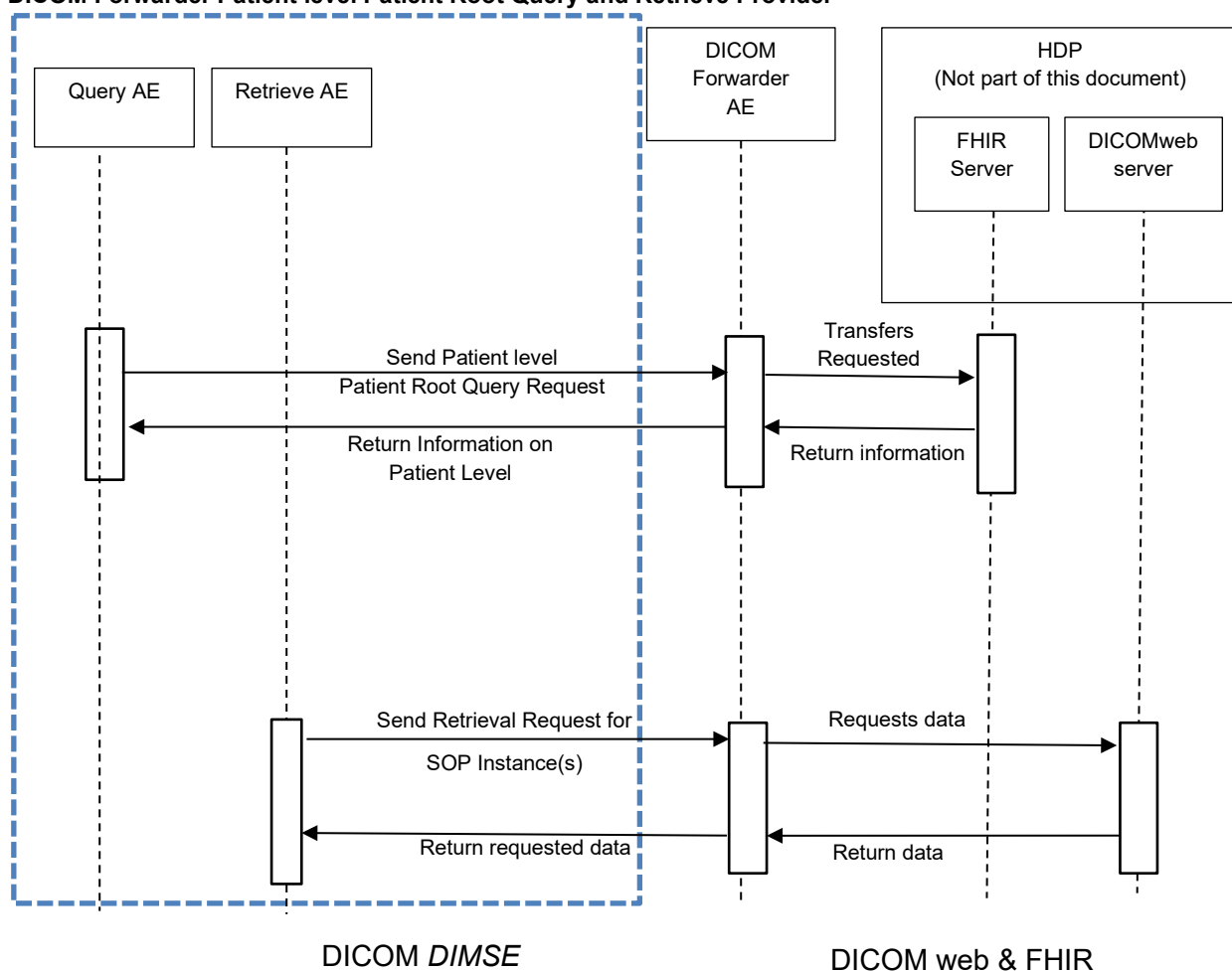
The blue mark indicates the device side of the communication.

Figure 4-4 DICOM Forwarder Study, Series and Instance level Query and Retrieve Provider



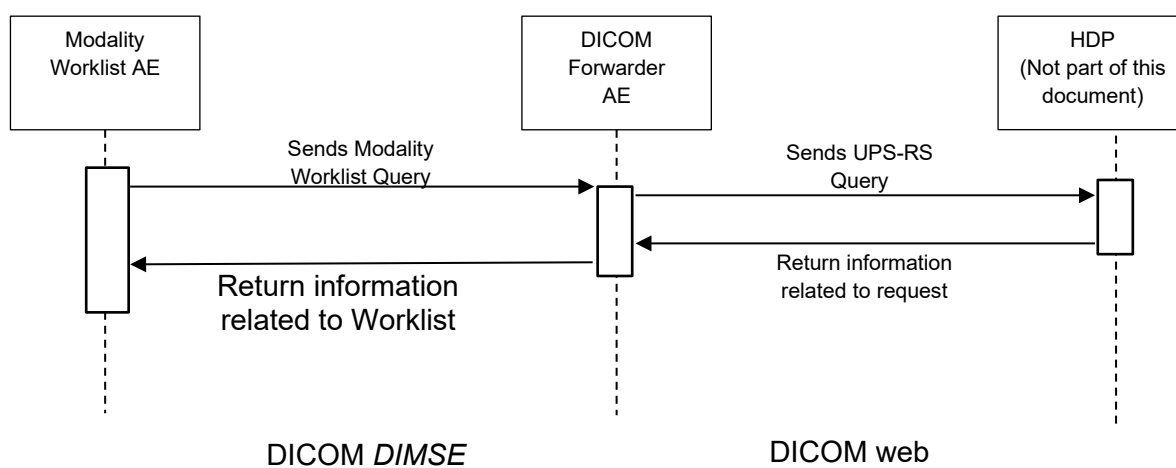
The blue mark indicates the device side of the communication.

Figure 4-5 DICOM Forwarder Patient level Patient Root Query and Retrieve Provider



The blue mark indicates the device side of the communication.

Figure 4-6 DICOM Forwarder Modality Worklist Provider



4.2 AE Specifications

4.2.1 DICOM Forwarder Application Entity Specification

4.2.1.1 SOP Classes

Table 4-1 SOP Classes for DICOM Forwarder AE

SOP Class Name	SOP Class UID	SCU	SCP
Verification	1.2.840.10008.1.1	No	Yes
Modality Worklist Information Model - FIND	1.2.840.10008.5.1.4.31	No	Yes
Ophthalmic Photography 8 Bit Image Storage	1.2.840.10008.5.1.4.1.1.77.1.5.1	Yes	Yes
Encapsulated PDF Storage	1.2.840.10008.5.1.4.1.1.104.1	Yes	Yes
Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7	Yes	Yes
Multi-frame Grayscale Byte Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7.2	Yes	Yes
Multi-frame True Color Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7.4	Yes	Yes
Raw Data Storage	1.2.840.10008.5.1.4.1.1.66	Yes	Yes
Visible Light Endoscopic Image Storage	1.2.840.10008.5.1.4.1.1.77.1.1	Yes	Yes
Visible Light Microscopic Image Storage	1.2.840.10008.5.1.4.1.1.77.1.2	Yes	Yes
Visible Light Photographic Image Storage	1.2.840.10008.5.1.4.1.1.77.1.4	Yes	Yes
Video Endoscopic Image Storage	1.2.840.10008.5.1.4.1.1.77.1.1.1	Yes	Yes
Video Microscopic Image Storage	1.2.840.10008.5.1.4.1.1.77.1.2.1	Yes	Yes
Video Photographic Image Storage	1.2.840.10008.5.1.4.1.1.77.1.4.1	Yes	Yes
Ophthalmic Tomography Image Storage	1.2.840.10008.5.1.4.1.1.77.1.5.4	Yes	Yes
Keratometry Measurements Storage	1.2.840.10008.5.1.4.1.1.78.3	Yes	Yes
Ophthalmic Axial Measurements Storage	1.2.840.10008.5.1.4.1.1.78.7	Yes	Yes
Intraocular Lens Calculations Storage	1.2.840.10008.5.1.4.1.1.78.8	Yes	Yes
Lensometry Measurements Storage	1.2.840.10008.5.1.4.1.1.78.1	Yes	Yes
Autorefracton Measurements Storage	1.2.840.10008.5.1.4.1.1.78.2	Yes	Yes
Subjective Refraction Measurements Storage	1.2.840.10008.5.1.4.1.1.78.4	Yes	Yes
Visual Acuity Measurements Storage	1.2.840.10008.5.1.4.1.1.78.5	Yes	Yes
Ophthalmic Visual Field Static Perimetry Measurements Storage	1.2.840.10008.5.1.4.1.1.80.1	Yes	Yes
Corneal Topography Map Storage	1.2.840.10008.5.1.4.1.1.82.1	Yes	Yes
Enhanced SR Storage	1.2.840.10008.5.1.4.1.1.88.22	Yes	Yes
Comprehensive SR Storage	1.2.840.10008.5.1.4.1.1.88.33	Yes	Yes
Grayscale Softcopy Presentation State Storage	1.2.840.10008.5.1.4.1.1.11.1	Yes	Yes
Key Object Selection Document Storage	1.2.840.10008.5.1.4.1.1.88.59	Yes	Yes
Ophthalmic Photography 16 Bit Image Storage	1.2.840.10008.5.1.4.1.1.77.1.5.2	Yes	Yes
Spectacle Prescription Report Storage	1.2.840.10008.5.1.4.1.1.78.6	Yes	Yes
Macular Grid Thickness and Volume Report Storage	1.2.840.10008.5.1.4.1.1.79.1	Yes	Yes

Ophthalmic Thickness Map Storage	1.2.840.10008.5.1.4.1.1.81.1	Yes	Yes
Wide Field Ophthalmic Photography Stereographic Projection Image Storage	1.2.840.10008.5.1.4.1.1.77.1.5.5	Yes	Yes
Wide Field Ophthalmic Photography 3D Coordinates Image Storage	1.2.840.10008.5.1.4.1.1.77.1.5.6	Yes	Yes
Ophthalmic Optical Coherence Tomography En Face Image Storage	1.2.840.10008.5.1.4.1.1.77.1.5.7	Yes	Yes
Ophthalmic Optical Coherence Tomography B-scan Volume Analysis Storage	1.2.840.10008.5.1.4.1.1.77.1.5.8	Yes	Yes
Surface Segmentation Storage	1.2.840.10008.5.1.4.1.1.66.5	Yes	Yes
Storage Commitment Push Model	1.2.840.10008.1.20.1	No	Yes
Patient Root Q/R Information Model – FIND	1.2.840.10008.5.1.4.1.2.1.1	No	Yes
Study Root Q/R Information Model – FIND	1.2.840.10008.5.1.4.1.2.2.1	No	Yes
Study Root Q/R Information Model – MOVE	1.2.840.10008.5.1.4.1.2.2.2	No	Yes

4.2.1.2 Associations Policies

4.2.1.2.1 General

The DICOM standard Application Context Name for DICOM 3.0 is always proposed:

Table 4-2 DICOM Application Context

Application Context Name	1.2.840.10008.3.1.1.1
--------------------------	-----------------------

4.2.1.2.2 Number of Associations

The DICOM Forwarder has one dedicated AE for all Storage, Query, Retrieve and Worklist related activities. The maximum number of associations is shown below.

Table 4-3 Number of associations

Maximum number of simultaneous associations	50
---	----

When the configured limit is exceeded, all further associations are rejected according to the reason documented in the table below.

Table 4-4 Association rejection reasons

Result	Source	Reason
2 - rejected-transient	c - DICOM UL service-provider (Presentation related function)	1 - temporary-congestion

4.2.1.2.3 Asynchronous Nature

DICOM Forwarder Application Software does not support asynchronous communication (multiple outstanding transactions over a single Association).

4.2.1.2.4 Implementation Identifying Information

Table 4-5 Number of associations

Implementation Class UID	1.2.276.0.75.2.5.140
Implementation Version Name	"DCMFWD-" + current application version, e.g., DCMFWD-1.3.0

4.2.1.3 Association Initiation Policy

4.2.1.3.1 Activity – User Requests Verification (SCU)

4.2.1.3.1.1 Description and Sequencing of Activities

The DICOM Forwarder AE sends DICOM Verification requests when the user requests a test of availability of a DICOM connection through the Web UI of Connectivity Module.

4.2.1.3.1.2 Proposed Presentation Contexts

Table 4-6 Presentation Contexts proposed by the DICOM Forwarder AE

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID List		
Verification	1.2.840.10008.1.1	ILE	1.2.840.10008.1.2	BOTH	No

4.2.1.3.2 Activity – DICOM Forwarder Issues Storage Request (SCU)

4.2.1.3.2.1 Description and Sequencing of Activities

The DICOM Forwarder AE will transfer request to the HDP Cloud and transmit instances as reaction to a received C-MOVE request. An association is established when the user initiates a transmit request. The DICOM Forwarder AE will transfer the request to the HDP Cloud and establish an association automatically in response to a C-MOVE request.

4.2.1.3.2.2 Proposed Presentation Contexts

The DICOM Forwarder AE may request any of the Presentation Contexts listed in the following table for Storage. The DICOM Forwarder AE will only propose the transfer syntax, which was used when the object was initially accepted by the HDP Cloud.

Table 4-7 Presentation Context Table

Presentation Context Table				
Abstract Syntax	Transfer Syntax		Role	Ext. Neg.
Name/UID	Name List	UID List		
All SOP classes listed in section "Transfer Image Storage SOP Classes"	Implicit VR Little Endian	1.2.840.10008.1.2	BOTH	No
	Explicit VR Little Endian	1.2.840.10008.1.2.1		
	Explicit VR Big Endian	1.2.840.10008.1.2.2		
	RLE Lossless	1.2.840.10008.1.2.5		
	JPEG Baseline	1.2.840.10008.1.2.4.50		
	JPEG Lossless	1.2.840.10008.1.2.4.70		
	JPEG 2000 (Lossless Only)	1.2.840.10008.1.2.4.90		
	JPEG 2000	1.2.840.10008.1.2.4.91		
All SOP classes listed in section "Transfer Video Image Storage SOP Classes"	MPEG2 MPML	1.2.840.10008.1.2.4.100	BOTH	No
	MPEG2 MPHL	1.2.840.10008.1.2.4.101		
	MPEG-4 AVC/H.264 High Profile / Level 4.1	1.2.840.10008.1.2.4.102		
	MPEG-4 AVC/H.264 BD-compatible High Profile / Level 4.1	1.2.840.10008.1.2.4.103		
	MPEG-4 AVC/H.264 High Profile / Level 4.2 For 2D Video	1.2.840.10008.1.2.4.104		
	MPEG-4 AVC/H.264 High Profile /	1.2.840.10008.1.2.4.105		

	Level 4.2 For 3D Video MPEG-4 AVC/H.264 Stereo High Profile / Level 4.2	1.2.840.10008.1.2.4.106		
All SOP classes listed in section "Transfer Other Storage SOP Classes"	Implicit VR Little Endian Explicit VR Little Endian Explicit VR Big Endian	1.2.840.10008.1.2 1.2.840.10008.1.2.1 1.2.840.10008.1.2.2	BOTH	No

4.2.1.3.2.3 SOP Specific Conformance – Store Objects

The DICOM Forwarder AE provides limited standard conformance to the DICOM Storage Service Class as an SCU: Neither it is guaranteed that the DICOM Default Transfer Syntax (Implicit VR Little Endian) is always proposed in a presentation context. Nor is an “on-the-fly” transcoding to the DICOM Default Transfer Syntax supported for instances received and stored in lossless compression. A successful C-Store response status will not generate any actions.

Table 4-8 STORE command return code mappings

Service Status	Further Meaning	Status Code	Behavior
Other than SUCCESS		B000	For C-MOVE Requests they will be reported in the C-STORE-RSP under "Number Of Failed Suboperations" (0000,1022)

Limitation of C-MOVE with large DICOM files:

Due to the limited temporary storage available as a **tmpfs** volume for retrieved DICOM files, it is not guaranteed that large DICOM files (i.e. those with file size over 100 MB) can be transferred successfully to the destination AE.

tmpfs (short for **Temporary File System**) is a temporary file storage in volatile memory instead of a persistent storage device. In case this storage volume allocated for DICOM Forwarder docker container gets filled with large retrieved DICOM file(s), a Storage Request will not be initiated by DICOM Forwarder to the destination AE.

4.2.1.3.3 Activity - Remote AE Requests Storage Commitment (SCU)

4.2.1.3.3.1 N-EVENT-Report

4.2.1.3.3.2 Description and Sequencing of Activities

The DICOM Forwarder AE will respond to N-ACTION requests, and initiate an N-EVENT-REPORT, based on the DICOM data availability in HDP via initiating a C-FIND Query on the DICOM WEB level. When the requested DICOM data(s) could be retrieved out of the HDP, then the Storage sequence had been closed, therefor the Storage Commitment is completed.

4.2.1.3.3.3 Proposed Presentation Contexts

The DICOM Forwarder AE may request the following Presentation Contexts listed in the following table.

Table 4-9 Proposed Presentation Contexts for Activity “Remote AE Requests Storage Commitment”

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID List		
Storage Commitment Push Model	1.2.840.10008.1.20.1	ILE	1.2.840.10008.1.2	SCP	No
		ELE	1.2.840.10008.1.2.1		
		EBE	1.2.840.10008.1.2.2		

4.2.1.3.3.4 SOP Specific Conformance – Storage Commitment

The DICOM Forwarder AE provides conformance to the DICOM Storage Commitment Service Class as an SCP.

The N-EVENT-REPORT contains the following Failure Reasons in the Failed SOP Sequence:

Table 4-10 Storage Commitment Failure Reasons

Service Status	Further Meaning	HDP Status Code	Failure Reason	Behavior
Failure	No such object instance	204	0112	The SOP instance is not available in HDP.
Failure	Processing failure	400	0110	Request is not processable/supported by HDP.
Failure	Refused: Not authorized	401	0124	Zeiss ID Authentication failed or invalid subscription key.
Failure	Forbidden	403	0110	Authorization error on HDP, e.g.: wrong tenant.
Failure	Refused: Not authorized	407	0124	Missing or invalid credentials are configured for http proxy server hence ZEISS ID token and/or HDP endpoints are not accessible.
Failure	Resource limitation	429	0213	Too many requests are sent to the HDP.
Failure	Internal server error	500	0110	The service is unavailable or busy.
Failure	Resource limitation	503	0213	The service is unavailable or busy.
Failure	Class instance conflict	200	0119	The SOP Class UID reported by HDP does not match with Referenced SOP Class UID included within N-ACTION request.
Failure	Duplicate SOP instances	200	0111	HDP reported multiple SOP Instances with the same SOP Instance UID.
Failure	Referenced SOP Class not supported		0122	The SOP Class is not supported by DICOM Forwarder.

4.2.1.4 Association Acceptance Policy

4.2.1.4.1 Activity – Remote AE Requests Verification (SCP)

4.2.1.4.1.1 Description and Sequencing of Activities

The DICOM Forwarder AE will respond to Verification requests from a Verification SCU. This way the requesting SCU can determine whether the respective DICOM Forwarder AEs are able to receive and handle incoming DICOM requests, including the end communication to/from HDP.

4.2.1.4.1.2 Accepted Presentation Contexts

Table 4-11 Acceptable Presentation Context for Activity “Remote AE Requests Verification”

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID List		
Verification	1.2.840.10008.1.1	ILE	1.2.840.10008.1.2	SCP	No

4.2.1.4.1.3 SOP Specific Conformance for Verification SOP Class as SCP

DICOM Forwarder provides standard conformance to the DICOM Verification Service Class as an SCP. One of the following status codes are returned:

Service Status	Further Meaning	Error Code	Behavior
Success	Success	0000	Operation performed properly.
Failure	Failure	A700	Out of Resource

DICOM Forwarder AE will always accept a Presentation Context for the Verification SOP Class with the default DICOM transfer syntax.

The DICOM DIMSE Verification includes the health API check answer returned by HDP, therefore if the HDP is not available, or has an issue, the DICOM DIMSE Verification answer contains this issue with an A700 Failure.

4.2.1.4.2 Activity - Remote AE issues Storage Request (SCP)

4.2.1.4.2.1 Description and Sequencing of Activities

The DICOM Forwarder AE will store DICOM instances that are sent to it from a remote SCU. All storage objects received by the DICOM Forwarder in a DICOM DIMSE protocol, are being transformed to DICOM WEB and sent towards the HDP.

4.2.1.4.2.2 Accepted Presentation Contexts

Table 4-12 Acceptable Presentation Context for Activity “Remote AE issues Storage Request”

Presentation Context Table				
Abstract Syntax		Transfer Syntax		Role
Name / UID		Name List	UID List	
All SOP classes listed in Table 1-1 under section “Transfer Image Storage SOP Classes”, “Transfer Video Image Storage SOP Classes” and “Transfer Other Storage SOP Classes”		The DICOM Forwarder will not limit any Transfer Syntax within the supported SOP Classes, and those will be transferred towards HDP		SCP
				No

4.2.1.4.2.3 SOP Specific Conformance – Store Objects

The DICOM Forwarder AE conforms to the DICOM Storage Service Class as a Level 2 (Full) SCP. No elements are discarded. Certain Attribute will be coerced, see 8.1.4. All Type 1, Type 2 and Type 3 attributes will be retained. Private attributes will be stored and included when the object is sent out later to a remote AE.

The DICOM Forwarder AE shall return the following status codes in response of a C-STORE Request:

Table 4-13 Storage status code mappings

Service Status	Further Meaning	Status Code	HDP Status Code	Behavior
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Ok	Success	0000	200	The SOP instance has been stored.
Ok	Accepted	0000	202 (*2)	The SOP instance has been stored with one or more of its optional attributes containing validation failure(s).
Error	Unable To Process	C000	204	Unexpected internal errors
Refused	Out Of Resources	A700	429	HDP rate limiting refuses connection.
Refused	Out Of Resources	A700	503	HDP Internal error, connection refused.
Error	Class instance conflict	A119	409	Class instance conflicts the ones HDP support
	Unable To Process	C000	415	The provided Content-Type is not supported
	Unable To Process	C000	406	The specified Accept header is not supported.
	Unable To Process	C000	400 (*1)	Unexpected internal errors.
Error	Refused: Not authorized	0124	401	Zeiss ID Authentication failed or invalid subscription key.
Error	Forbidden	C000	403	HDP Core reported authorization error.
Error	Refused: Not authorized	0124	407	Missing or invalid credentials are configured for http proxy server hence ZEISS ID token and/or HDP endpoints are not accessible.
Error	Unable To Process	0111	409 – Reason: 45070	SOP Instance UID already exists
Error	Unable To Process	0110	409 – Reason: 272	Processing failure
Error	Unable To Process	0110	409 – Reason: 43264	Processing failure
Error	Unable To Process	C000	409 – Reason: 43265	Unexpected internal errors
Error	Unable To Process	C000	409 – Reason: 45071	Unexpected internal errors
Error	Internal Server error	0110	500	HDP Core reported that an unexpected error occurred.
Success	Success	0000	200	Operation performed properly.

Remark (*1):

The following attributes (see Table 4-14 Required attributes validated upon storage) are required to be present in every instance attempted to be stored. The value of these attributes is validated by the DICOMweb server according to the attribute's VR type. In case the validation fails, storage of the instance is rejected and HDP returns an http status code of 400. This is then mapped to Unable To Process (C000) status code which is returned by the DICOM Forwarder AE to the remote AE.

Table 4-14 Required attributes validated upon storage

Description	Tag
Study Instance UID	(0020,000D)

Series Instance UID	(0020,000E)
SOP Instance UID	(0008,0018)
SOP Class UID	(0008,0016)
Patient ID	(0010,0020)

Remark (*2):

If present in the DICOM instance, the DICOMweb server performs validation also for the following queryable attributes (see Table 4-15 Optional attributes validated upon storage) according to the attribute's VR type when an instance is attempted to be stored. However, in case the validation of one or more of these attributes fail, the instance is not rejected but is stored with the original attributes, even if those attributes failed validation. HDP returns an http status code of 202 in this case which is then mapped to status code Success (0000) and is returned by the DICOM Forwarder AE to the remote AE.

Table 4-15 Optional attributes validated upon storage

Description	Tag
Patient's Name	(0010,0010)
Patient's Birth Date	(0010,0030)
Issuer of Patient ID	(0010,0021)
Patient Sex	(0010,0040)
Study ID	(0020,0010)
Accession Number	(0008,0050)
Study Date	(0008,0020)
Study Description	(0008,1030)
Referring Physician's Name	(0008,0090)
Modality	(0008,0060)
Manufacturer's Model Name	(0008,1090)
Acquisition Date Time	(0008,002A)
Study Time	(0008,0030)
Series Number	(0020,0011)
Instance Number	(0020,0013)
Performed Procedure Step Start Date	(0040,0244)

If a store request results in validation failure(s) for one or more of the above listed optional attributes listed in Table 4-15, subsequent queries are affected by the following notable behaviors:

- In general, SOP instances with optional attribute validation failures are considered in subsequent queries based on attribute(s) with valid values and so the query result contains matching study/series/instance.
- However, the affected attributes with validation failures are only included in the query response with empty value.
- Any SOP instance that produced a validation failure is not considered in queries based on attribute(s) with validation failure(s) and so the query response may provide incomplete search results.

- However, optional Study/Series level queryable attribute(s) of SOP instances that failed validation are included in Study/Series level query response datasets if the incorrect values are overwritten by storing instances for the same study/series with correct values after the failed one or if the values are already stored correctly by a previous instance.

Retrieving a study/series/instance always returns the original binary DICOM data with the original attributes, even if those attributes failed validation.

Limitation of C-STORE with large DICOM files:

Due to the limited temporary storage and potentially lower upload bandwidth, all the files above a configured threshold are uploaded to the HDP after the C-STORE association is closed with a 0000 – Success message, even if the upload is not complete. If an error happens during such an upload, the DICOM Forwarder tries to re-upload the file multiple times, after that it will be deleted without archiving it. The file size threshold and the number of retries can be configured implicitly via the selection of network profile settings in the Web UI of Connectivity Module.

In order to prevent a possible data loss, the device shall utilize a Storage Commitment request before data deletion, to confirm if the data had been archived successfully or not.

4.2.1.4.3 Activity – Remote AE issues Patient or Study Root Query (SCP)

4.2.1.4.3.1 Description and Sequencing of Activities

The DICOM Forwarder AE will respond to query requests that are sent to it by a remote SCU.

4.2.1.4.3.2 Accepted Presentation Contexts

Table 4-16 Acceptable Presentation Context

Presentation Context Table				
Abstract Syntax	Transfer Syntax		Role	Ext. Neg.
Name / UID	Name List	UID List		
Patient Root Q/R Information Model – FIND 1.2.840.10008.5.1.4.1.2.1.1	ILE	1.2.840.10008.1.2	SCP	YES
Study Root Q/R Information Model – FIND - 1.2.840.10008.5.1.4.1.2.2.1	ILE	1.2.840.10008.1.2	SCP	YES

C-Find Extended Negotiation is supported: For the field "Relational-queries", the value "1" is returned, meaning that those are supported by the DF AE.

4.2.1.4.3.3 SOP Specific Conformance – Find Objects

The DICOM Forwarder AE provides standard conformance to the DICOM Query/Retrieve Service Class as an SCP. The DICOM Forwarder AE supports both Hierarchical Queries and the Relational-queries extended SCP behavior. Matching for attributes with VR PN (Person Name) is always done case-insensitive. The following tables contain detailed information on matching and return keys.

Table 4-17 Matching Types

Abbreviation	Description
U	Universal Matching
S	Single Value Matching
*	Wildcard Matching

R	Range Matching
UNIQUE	Unique query key for the respective level

Table 4-18 Patient Root / Patient Level Attributes

Description	Tag	Matching Type
Patient's Name	(0010,0010)	U S *
Patient ID	(0010,0020)	U S
Patient's Birth Date	(0010,0030)	U S R
Issuer of Patient ID	(0010,0021)	U S
Patient Sex	(0010,0040)	U S
Ethnic Group	(0010,2160)	U
Patient Comments	(0010,4000)	U
Other Patient IDs	(0010,1000)	U
Other Patient IDs Sequence	(0010,1002)	U
›Patient ID	(0010,0020)	U
›Issuer of Patient ID	(0010,0021)	U

Table 4-19 Patient Root / Study Level Attributes

Description	Tag	Matching Type
Study Instance UID	(0020,000D)	U S, UNIQUE
Study ID	(0020,0010)	U
Accession Number	(0008,0050)	U S
Study Date	(0008,0020)	U S R
Referring Physician's Name	(0008,0090)	U S *
Modalities in Study (*1)	(0008,0061)	U S
Study Time	(0008,0030)	U
Study Description	(0008,1030)	U
Admitting Diagnoses Description	(0008,1080)	U
Instance Availability	(0008, 0056)	U

Remark (*1):

"Modalities in Study" is supported ONLY on STUDY LEVEL and not on any other levels (i.e. SERIES or INSTANCE levels). This attribute is only supported in Single Value Match query match type. In case multiple values are provided, only the first value is used for filtering.

Table 4-20 Study Root / Study Level Attributes

Description	Tag	Matching Type
Patient's Name	(0010,0010)	U S *
Patient ID	(0010,0020)	U S
Patient's Birth Date	(0010,0030)	U S R
Issuer of Patient ID	(0010,0021)	U S
Patient Sex	(0010,0040)	U S
Ethnic Group	(0010,2160)	U
Patient Comments	(0010,4000)	U
Other Patient IDs	(0010,1000)	U
Patient's Birth Time	(0010, 0032)	U
Study Instance UID	(0020,000D)	U S, UNIQUE
Study ID	(0020,0010)	U
Accession Number	(0008,0050)	U S
Study Date	(0008,0020)	U S R
Referring Physician's Name	(0008,0090)	U S *
Modalities in Study (*1)	(0008,0061)	U S
Study Time	(0008,0030)	U
Study Description	(0008,1030)	U
Admitting Diagnoses Description	(0008,1080)	U
Instance Availability	(0008, 0056)	U

Remark (*1):

"Modalities in Study" is supported ONLY on STUDY LEVEL and not on any other levels (i.e. SERIES or INSTANCE levels). This attribute is only supported in Single Value Match query match type. In case multiple values are provided, only the first value is used for filtering.

Table 4-21 Series Level Attributes

Description	Tag	Matching Type
Modality	(0008,0060)	U S
Series Instance UID	(0020,000E)	U S, UNIQUE

Manufacturer's Model Name	(0008,1090)	U S
Performing Physicians' Name	(0008,1050)	U
Series Description	(0008,103E)	U
Performed Procedure Step Start Date	(0040,0244)	U
Performed Procedure Step Start Time	(0040,0245)	U
Request Attributes Sequence	(0040,0275)	U
Requested Procedure ID	(0040,1001)	U
Scheduled Procedure Step ID	(0040,0009)	U
Series Number	(0020,0011)	U
Laterality	(0020,0060)	U
Series Date	(0008,0021)	U
Series Time	(0008,0031)	U

Table 4-22 Instance Level Attributes

Description	Tag	Matching Type
SOP Instance UID	(0008,0018)	U S, UNIQUE
Acquisition Date Time	(0008,002A)	U S R
SOP Class UID	(0008,0016)	U S
Instance Creation Date	(0008,0012)	U
Instance Creation Time	(0008,0013)	U
Instance Number	(0020,0013)	U
Referenced Instance Sequence	(0008,114A)	U
Referenced SOP Instance UID	(0008,1155)	U
Referenced SOP Class UID	(0008,1150)	U
Document Title	(0042,0010)	U
Image Laterality (*1)	(0020,0062)	U
Measurement Laterality	(0024,0113)	U
Rows	(0028,0010)	U
Columns	(0028,0011)	U

Bits Allocated	(0028,0100)	U
Number of Frames	(0028,0008)	U
Instance Availability	(0008,0056)	U
Image Type	(0008,0008)	U

Remark (*1):

In case the optional *Image Laterality* (0020,0062) attribute value is requested in a Patient or Study Root Query but no value was stored for the attribute, it's value is attempted to be set first based on *Measurement Laterality* (0024,0113) – or if that is also empty – based on the Series level *Laterality* (0020,0060) attribute value if any of the latter attributes are included in the C-FIND request dataset.

Table 4-23 Private attributes - Instance Level

Description	Tag	Matching Type
Pattern Type	(0405,xx01) VR=CS, VM=1	U S
Signal Strength	(0405,xx1A) VR=FL, VM=1	U
Number of pixels in Y dimension of B scan (in OCT Cube Sequence)	(0407,xxA1) VR=SQ, VM=1 > (0407,xx01) VR=US, VM=1	U
Number of pixels in X dimension of B scan (in OCT Cube Sequence)	(0407,xxA1) VR=SQ, VM=1 > (0407,xx02) VR=US, VM=1	U
Number of B scans in acquisition (in OCT Cube Sequence)	(0407,xxA1) VR=SQ, VM=1 > (0407,xx03) VR=US, VM=1	U

The DICOM Forwarder AE shall return the following status codes in response of a C-FIND Request.

Table 4-24 Query command Status Codes

Service Status	Further Meaning	Status Code	HDP Status Code	Behavior
Success	Success	FF00	200	The search completed successfully. Response contains matching resource, matches are continuing.
Success	Success	FF01	200	The search completed successfully, but one or more optional query match keys received in the query request were ignored. Response contains matching resource, matches are continuing.
Success	Success	0000	200	The search completed successfully. All matching resources returned; matching is complete.
Success	No Content	0000	204	No Content
Error	Bad request	C000	400	HDP Core reported that the request was badly formatted. Possible DICOM Forwarder Issue.
Error	Refused: Not authorized	0124	401	Zeiss ID Authentication failed or invalid subscription key.
Error	Forbidden	C000	403	HDP Core reported authorization error.
Error	Refused: Not authorized	0124	407	Missing or invalid credentials are configured for http proxy server hence ZEISS ID token and/or HDP endpoints are not accessible.

Error	Resource limitation	A700	429	HDP Core reported too many requests
Error	Internal Server error	0110	500	HDP Core reported that an unexpected error occurred.
Refused	Out Of Resources	A700	503	HDP Internal error, connection refused.

Limitation of C-FIND Study and Patient Root Queries with large number of matches:

In order to avoid potential timeout issues for the established DICOM DIMSE associations for C-FIND operations, the maximum number of results is limited to 200 patients, studies, series or instances for queries performed at the corresponding levels.

This number is set by taking into account the maximum allowed size for an http response by the DICOMweb server (in case of STUDY, SERIES or IMAGE level Study and Patient Root Queries) and by the FHIR server (in case of PATIENT level Patient Root Queries) hosted by HDP.

No errors are reported by DICOM Forwarder in case the number of matches exceeded this value, but only the first 200 results are returned.

4.2.1.4.4 Activity – Remote AE Requests Storage Commitment (SCP)

4.2.1.4.4.1 N-ACTION-RSP

4.2.1.4.4.2 Description and Sequencing of Activities

When DICOM Forwarder receives a storage commitment request, it checks whether the requested SOP instances are present in HDP and sends an N-EVENT-REPORT-RQ. DICOM Forwarder accepts a list of references to one or more DICOM SOP instances.

4.2.1.4.4.3 Accepted Presentation Contexts

Table 4-25 Acceptable Presentation Context for Activity “Remote AE Requests Storage Commitment”
Presentation Context Table

Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID 1.2.840.10008. ...	Name List	UID List 1.2.840.10008. ...		
Storage Commitment Push Model	1.2.840.10008.1.20.1	ILE	1.2.840.10008.1.2	SCP	No

4.2.1.4.4.4 SOP Specific Conformance – Storage Commitment

The DICOM Forwarder AE provides conformance to the DICOM Storage Commitment Service Class as an SCP. The number of referenced instances within one request is not limited. The N-EVENT-REPORT message is always sent in a separate association from the N-ACTION operation.

For Storage Commitment operations, the DICOM Forwarder AE will only accept association requests for known (configured in HDP) remote AEs.

One of the following Status codes is returned by the DICOM Forwarder AE in an N-ACTION response:

Table 4-26 Storage Commitment N-ACTION-RSP Status Codes

Service Status	Further Meaning	Status Code	HDP Status Code	Behavior
Failure	Processing failure	0110	400	Request is not processable/supported by HDP.

Success	Success	0000	200	Operation performed properly
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4.2.1.4.5 Activity – Remote AE Issues Retrieve Command (SCP)

4.2.1.4.5.1 Description and Sequencing of Activities

The DICOM Forwarder AE will respond to C-MOVE requests that are sent to it by a remote SCU. The DICOM Forwarder AE will establish a new association with the remote AE specified in the move destination for the C-STORE sub operation. The DICOM Forwarder AE will always only propose the transfer syntax used when the object was initially sent to the HDP.

4.2.1.4.5.2 Accepted Presentation Contexts

Table 4-27 Acceptable Presentation Context for Activity “Remote AE Issues Retrieve Command”

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID List		
Study Root Q/R Information Model – MOVE	1.2.840.10008.5.1.4.1.2.2.2	ILE	1.2.840.10008.1.2	SCP	Yes

C-MOVE Extended Negotiation is supported: For the field “Relational-retrieve”, the value “1” is returned, meaning that those are supported by the DF AE.

4.2.1.4.5.3 SOP Specific Conformance – Move Objects

The DICOM Forwarder AE will establish an association with the destination specified in the C-MOVE request. One or more of the Presentation Contexts listed in chapter 4.2.1.4.2 “Activity – Remote AE issues Storage Request (SCP)” may be negotiated in this association. The following tables contain detailed information on supported matching keys:

Table 4-28 Study Root / Study Level Attributes

Description	Tag	Matching Type
Study Instance UID	(0020,000D)	List of UID Matching

Table 4-29 Series Level Attributes

Description	Tag	Matching Type
Series Instance UID	(0020,000E)	List of UID Matching

Table 4-30 Instance Level Attributes

Description	Tag	Matching Type
SOP Instance UID	(0008,0018)	List of UID Matching

In case (default) hierarchical retrieve mode (i.e. when “Relational-retrieve” mode is not negotiated), a list of UIDs are only supported on the given Query/Retrieve Level while UIDs from all „above“ levels must also be provided in Single Value matching mode:

- For STUDY level retrieve, a list of UIDs via Study Instance UID attribute.
- For SERIES level retrieve, a list of UIDs via Series Instance UID attribute and a single value via Study Instance UID attribute for the (single) corresponding Study.
- For IMAGE level retrieve, a list of UIDs via SOP Instance UID attribute and a single value via Series Instance UID and Study Instance UID attributes of the corresponding (single) Series and (single) Study.

In case “Relational-retrieve” mode is negotiated via Extended Negotiation, a list of UIDs are supported on the given Query/Retrieve Level:

- For STUDY level retrieve, a list of UIDs via Study Instance UID attribute.
- For SERIES level retrieve, a list of UIDs via Series Instance UID attribute.
- For IMAGE level retrieve, a list of UIDs via SOP Instance UID attribute.

In case lenient retrieve mode is enabled (which is the default setting) DICOM Forwarder leniently accepts C-MOVE request without requiring unique keys for all levels above the current Query/Retrieve Level even without negotiating relational mode during association initialization. Only the same UIDs must be present in the request dataset as defined in “Relational-retrieve” mode based on the given Query/Retrieve Level.

The DICOM Forwarder AE shall return the following status codes in response of a C-MOVE Request:

Table 4-31 Retrieve command Status Codes

Service Status	Further Meaning	Status Code	HDP Status Code	Behavior
Success		0000	200	The operation is successful.
Failure	Unable to process	C000	400	HDP Core reported that the request was badly formatted.
Failure	Refused: Not authorized	0124	401	Authentication token is expired, or subscription key invalid.
Failure	Forbidden	C000	403	HDP Core reported authorization error.
Failure	No such object instance	0112	404	The specified DICOM resource could not be found.
Failure	Unable to process	C000	406	Communication rejected by HDP, a possible DICOM Forwarder issue.
Failure	Refused: Not authorized	0124	407	Missing or invalid credentials are configured for http proxy server hence ZEISS ID token and/or HDP endpoints are not accessible.
Failure	Resource limitation	A700	429	Too many requests are sent to the HDP.
Failure	Internal Server error	0110	500	HDP Core reported that an unexpected error occurred.
Failure	Resource limitation	A702	503	The service is unavailable or busy.

4.2.1.4.6 Activity – Remote AE sends MWL Query (SCP)

4.2.1.4.6.1 Description and Sequencing of Activities

The DICOM Forwarder MWL AE will respond to DICOM C-FIND requests in response to a remote AE querying for a Modality Worklist.

4.2.1.4.6.2 Accepted Presentation Context

Table 4-32 Acceptable Presentation Context for Activity “Remote AE sends MWL Query”

Presentation Context Table				
Abstract Syntax		Transfer Syntax		Role
Name	UID	Name List	UID List	

Modality Worklist Information Model - FIND	1.2.840.10008.5.1.4. 31	ILE	1.2.840.10008.1.2	SCP	Yes
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4.2.1.4.6.3 SOP Specific Conformance for Modality Worklist SOP Class as SCP

The DICOM Forwarder MWL AE provides standard conformance to the DICOM Basic Worklist Management Service Class.

Table 4-33 Matching Types

Abbreviation	Description
U	Universal Matching
S	Single Value Matching
*	Wildcard Matching
R	Range Matching

The following table contains detailed information on matching keys:

Table 4-34 Modality Worklist - Matching Keys

Description	Tag	Matching Type
Accession Number	(0008,0050)	U S
Requested Procedure ID	(0040,1001)	U S
Scheduled Procedure Step Sequence	(0040,0100)	
›Scheduled Station AE Title	(0040,0001)	U S
›Scheduled Procedure Step Start Date	(0040,0002)	U R
›Modality	(0008,0060)	U S
Patient's Name	(0010,0010)	U S *
Patient ID	(0010,0020)	U S

The following table contains the supported return keys:

Table 4-35 Modality Worklist - Return Keys

Description	Tag
Scheduled Procedure Step Sequence	(0040,0100)
>Scheduled Station AE Title	(0040,0001)
>Scheduled Procedure Step Start Date	(0040,0002)
>Scheduled Procedure Step Start Time	(0040,0003)
>Scheduled Performing Physician Name	(0040,0006)
>Scheduled Procedure Step Location	(0040,0011)
>Scheduled Procedure Step Status	(0040,0020)
>Scheduled Protocol Code Sequence	(0040,0008)
>> Code Meaning	(0008,0104)
>> Code Value	(0008,0100)

>> Coding Scheme Designator	(0008,0102)
>> Coding Scheme Version	(0008,0103)
>Modality	(0008,0060)
>Scheduled Procedure Step Description	(0040,0007)
>Scheduled Procedure Step ID	(0040,0009)
Requested Procedure ID	(0040,1001)
Requested Procedure Comments	(0040,1400)
Requested Procedure Description	(0032,1060)
Study Instance UID	(0020,000D)
Study Date (=Scheduled Procedure Step Start Date)	(0008,0020)
Study Time (=Scheduled Procedure Step Start Time)	(0008,0030)
Accession Number	(0008,0050)
Requested Procedure Code Sequence (=Scheduled Protocol Code Sequence)	(0032,1064)
> Code Meaning	(0008,0104)
> Code Value	(0008,0100)
> Coding Scheme Designator	(0008,0102)
> Coding Scheme Version	(0008,0103)
Referring Physician's Name	(0008,0090)
Patient's Name	(0010,0010)
Patient ID	(0010,0020)
Issuer of Patient ID	(0010,0021)
Patient's Birth Date	(0010,0030)
Patient's Sex	(0010,0040)
Specific Character Set	(0008,0005)
Patient Comments	(0010,4000)
Requesting Physician	(0032,1032)
Allergies	(0010,2110)
Ethnic Group	(0010,2160)

Regarding the DICOM DIMSE and DICOM Web protocol change, a mapping is used.

Table 4-36 MWL and UPS query match key mappings

MWL query match key attribute name	MWL query match key attribute id	Search type	Mapped UPS-RS query match key attribute name	Mapped UPS query match key attribute id
Scheduled Procedure Step Sequence	(0040,0100)		-	
-	-		Scheduled Station Name Code Sequence (*2)	(0040,4025)
>Scheduled Station AE Title (*1)	(0040,0001)	U S	>Code Value	(0008,0100)
›Scheduled Procedure Step Start Date	(0040,0002)	U R	Scheduled Procedure Step Start Date Time	(0040,4005)
-	-		Scheduled Station Class Code Sequence (*3)	(0040,4026)
>Modality	(0008,0060)	U S	>Code Value	(0008,0100)
-	-		Referenced Request Sequence	(0040, A370)
Requested Procedure ID	(0040,1001)	U S	>Requested Procedure ID	(0040,1001)
Accession Number	(0008,0050)	U S	›Accession Number	(0008,0050)
Patient's Name	(0010,0010)	U S *	Patient's Name	(0010,0010)
Patient ID	(0010,0020)	U S	Patient ID	(0010,0020)
-	-	Predefined value "SCHEDULED"	Procedure Step State = SCHEDULED	(0074,1000)

Remark (*1):

The MWL attribute *Scheduled Station AE Title (0040,0001)* under the *Scheduled Procedure Step Sequence (0040,0100)* is mapped to the UPS query match key attribute *Code Value* under the *Scheduled Station Name Code Sequence (0040,4025)* based on the connected device configuration.

- If the *Scheduled Station AE Title* query match key value is not present in the C-FIND RQ dataset, then no mapping is performed for this attribute by DICOM Forwarder, i.e. *Code Value* under *Scheduled Station Name Code Sequence* is not used in UPS-RS search transaction as a query match key attribute value.
- If the *Scheduled Station AE Title* value is included as a query match key in the MWL C-FIND RQ dataset but it references and unknown or inactive device, empty MWL C-FIND response dataset is returned to the caller AE.
- If the *Scheduled Station AE Title* value is included as a query match key in the MWL C-FIND RQ dataset and it references a configured and active device, then the configured SNC (Station Name Code) value of the device is looked up based on the specified *Scheduled Station AE Title* value and is used in the UPS-RS search transaction as the *Code Value* under *Scheduled Station Name Code Sequence* query match key attribute value.

Remark (*2):

For the *Scheduled Station Name Code Sequence (0040,4025)* of an UPS worklist item besides the *Code Value (0008, 0100)* attribute there are some other Type 1 – Mandatory fields that must be present within the stored worklist item e.g. *Code Scheme Designator (0008, 0102)* and *Code Meaning (0008, 0104)*. But as the *Scheduled Station AE Title (0040,0001)* MWL attribute value only gets mapped to the *Code Value*, these other mandatory fields are not present in the current MWL to UPS mapping table.

Remark (*3):

For the *Scheduled Station Class Code Sequence (0040,4026)* of an UPS worklist item besides the *Code Value (0008, 0100)* attribute there are some other Type 1 – Mandatory fields that must be present within the stored worklist item e.g. *Code Scheme Designator (0008, 0102)* and *Code Meaning (0008, 0104)*. But as the *Modality (0008,0060)* MWL attribute value only gets mapped to the *Code Value*, these other mandatory fields are not present in the current MWL to UPS mapping table.

Table 4-37 UPS to MWL query return key mappings

UPS-RS query return key attribute name	UPS-RS query return key attribute id	Mapped MWL query return key attribute name	Mapped MWL query return key attribute id
Scheduled Station Name Code Sequence	(0040,4025)	-	
-	-	Scheduled Procedure Step Sequence	(0040,0100)
>Code Value (*1)	(0008,0100)	>Scheduled Station AE Title	(0040,0001)
Scheduled Procedure Step Start Date Time (*2)	(0040,4005)	>Scheduled Procedure Step Start Date	(0040,0002)
Scheduled Procedure Step Start Date Time (*2)	(0040,4005)	Study Date	(0008,0020)
Scheduled Procedure Step Start Date Time (*3)	(0040,4005)	>Scheduled Procedure Step Start Time	(0040,0003)
Scheduled Procedure Step Start Date Time (*3)	(0040,4005)	Study Time	(0008,0030)
Scheduled Station Class Code Sequence	(0040,4026)	-	-
>Code Value (*4)	(0008,0100)	>Modality	(0008,0060)
Procedure Step Label	(0074,1204)	>Scheduled Procedure Step Description	(0040, 0007)
Scheduled Workitem Code Sequence (*5)	(0040,4018)	>Scheduled Protocol Code Sequence	(0040, 0020)
>Code Value	(0008,0100)	>> Code Value	(0008,0100)
>Coding Scheme Designator	(0008,0102)	>> Coding Scheme Designator	(0008,0102)
>Coding Scheme Version	(0008,0103)	>> Coding Scheme Version	(0008,0103)
>Coding Meaning	(0008,0104)	>> Coding Meaning	(0008,0104)
Referenced Request Sequence (*6)	(0040,A370)	-	-
>Requested Procedure ID (*7)	(0040,1001)	Scheduled Procedure Step ID	(0040,0009)
>Requested Procedure ID (*7)	(0040,1001)	Requested Procedure ID	(0040,1001)
>Requested Procedure Description	(0032,1060)	Requested Procedure Description	(0032,1060)
>Requested Procedure Comments	(0040,1400)	Requested Procedure Comments	(0040,1400)
>Study Instance UID	(0020,000D)	Study Instance UID	(0020,000D)
>Accession Number	(0008,0050)	Accession Number	(0008,0050)

>Referring Physician's Name	(0008,0090)	Referring Physician's Name	(0008,0090)
>Requesting Physician	(0032,1032)	Requesting Physician	(0032,1032)
>Requested Procedure Code Sequence	(0032,1064)	Requested Procedure Code Sequence	(0032,1064)
>>Code Value	(0008,0100)	>Code Value	(0008,0100)
>>Coding Scheme Designator	(0008,0102)	>Coding Scheme Designator	(0008,0102)
>>Coding Scheme Version	(0008,0103)	> Coding Scheme Version	(0008,0103)
>>Coding Meaning	(0008,0104)	> Coding Meaning	(0008,0104)
Patient's Name	(0010,0010)	Patient's Name	(0010,0010)
Patient ID	(0010,0020)	Patient ID	(0010,0020)
Issuer of Patient ID	(0010,0021)	Issuer of Patient ID	(0010,0021)
Patient's Birth Date	(0010,0030)	Patient's Birth Date	(0010,0030)
Patient's Sex	(0010,0040)	Patient's Sex	(0010,0040)
Patient Comments	(0010,4000)	Patient Comments	(0010,4000)
Allergies	(0010,2110)	Allergies	(0010,2110)
Ethnic Group	(0010,2160)	Ethnic Group	(0010,2160)

Remark (*1):

The *Code Value* under the sequence *Scheduled Station Name Code Sequence (0040,4025)* of the UPS workitem is mapped to the *Scheduled Station AE Title (0040,0001)* attribute under *Scheduled Procedure Step Sequence (0040,0100)* in the MWL worklist based on the connected device configuration by selecting an associated device. This selection can be necessary due to the following reasons:

- The *Scheduled Station Name Code Sequence (0040,4025)* in a UPS workitem can contain multiple items referencing multiple device(s).
- The *Scheduled Station Name Code Sequence (0040,4025)* in a UPS workitem can also be empty (i.e. not explicitly referencing any device).
- The related *Scheduled Station AE Title (0040,0001)* attribute under *Scheduled Procedure Step Sequence (0040,0100)* in the MWL worklist is a *Type1* mandatory attribute and a *single* value is required to be returned in the C-FIND RSP dataset for this attribute.

DICOM Forwarder omits SNC (Station Name Code) value(s) specified under a UPS workitem or the whole UPS workitem itself entirely from mapping to MWL C-FIND RSP dataset(s) in the following cases:

1. If a UPS workitem contains such SNC (Station Name Code) value(s) that is/are unknown to DICOM Forwarder (i.e. no matching connected device is available with matching SNC value) or the device(s) with matching SNC value(s) is/are inactive, those SNC values is/are omitted from mapping and so no corresponding *Scheduled Station AE Title* value is included in MWL C-FIND RSP dataset(s).
2. If a UPS workitem contains ONLY such SNC (Station Name Code) value(s) that are unknown to DICOM Forwarder or ALL the SNC value(s) is/are referring to inactive connected device(s), then the UPS workitem is altogether omitted from the C-FIND RSP dataset(s) with all of its attributes.

In case *Scheduled Station Name Code Sequence* in the UPS workitem contains at least one such SNC value that references a configured and active device, then a single *Scheduled Station AE Title* value of an associated device is selected by DICOM Forwarder from the list of configured and active connected devices based on the following rules and according to the following order:

1. In case there is a single SNC value under *Scheduled Station Name Code Sequence* in the UPS workitem, the related AE Title is fetched by DICOM Forwarder from the list of known and active connected devices for the corresponding device and this value is included in *Scheduled Station AE Title* attribute under the *Scheduled Procedure Step Sequence* of the MWL C-FIND RSP dataset.
2. In case there are multiple SNC values under the *Scheduled Station Name Code Sequence* in the UPS workitem, a value-selection logic is applied by DICOM Forwarder according to the following rules:
 - a. DICOM Forwarder selects the same value as the *Scheduled Station AE Title* query match key value in MWL-C-FIND RQ dataset and includes it also as *Scheduled Station AE Title* attribute value in the MWL C-FIND RSP dataset in case MWL-C-FIND RQ dataset contains a non-empty *Scheduled Station AE Title* query match key.
 - b. If *Scheduled Station AE Title* query match key value is not present in the MWL-C-FIND RQ dataset then DICOM Forwarder selects the SNC value that maps to the AE Title of the (known and active) connected device of the MWL-C-FIND RQ association initiator DICOM client.
 - c. If none of the AE Titles mapped from (multiple) *Scheduled Station Name Code Sequence* values match the AE Title of the MWL-C-FIND RQ association initiator DICOM device, the first such SNC Value from the *Scheduled Station Name Code Sequence* is mapped to the *Scheduled Station AE Title* in the MWL C-FIND RSP dataset that references a known and active device.
3. In case there are no SNC values under the *Scheduled Station Name Code Sequence* in the UPS workitem, DICOM Forwarders assigns a value to the *Scheduled Station AE Title* in the MWL C-FIND RSP according to the following logic:
 - a. If the MWL-C-FIND RQ dataset contains *Scheduled Station AE Title* as a query match key value (which refers to a known and active device), DICOM Forwarder assigns the same value to the *Scheduled Station AE Title* attribute in the MWL C-FIND RSP dataset also for the corresponding MWL worklist.
 - b. If *Scheduled Station AE Title* query match key value is not present in the MWL-C-FIND RQ dataset then DICOM Forwarder assigns the (known and active) AE Title of the MWL-C-FIND RQ association initiator DICOM client to the *Scheduled Station AE Title* attribute in the MWL C-FIND RSP dataset.

Remark (*2):

Date value extracted from *Scheduled Procedure Step Start Date Time (0040,4005)* attribute of UPS workitem is mapped to both *Study Date (0008,0020)* and *Scheduled Procedure Step Start Date (0040,0003)* under *Scheduled Procedure Step Sequence (0040,0100)* of the related MWL worklist.

Remark (*3):

Time value extracted from *Scheduled Procedure Step Start Date Time (0040,4005)* attribute of UPS workitem is mapped to both *Study Time (0008,0030)* and *Scheduled Procedure Step Start Time (0040,0002)* under *Scheduled Procedure Step Sequence (0040,0100)* of the related MWL worklist.

Remark (*4):

For the *Scheduled Station Class Code Sequence (0040,4026)* of an UPS worklist item besides the *Code Value (0008, 0100)* attribute there are some other Type 1 – Mandatory fields that must be present within the stored worklist item e.g. *Code Scheme Designator (0008, 0102)* and *Code Meaning (0008, 0104)*. But as the *Modality (0008,0060)* MWL attribute value only gets mapped to the *Code Value*, these other mandatory fields are not present in the current MWL to UPS mapping table.

Because the VM (Value Multiplicity) of *Modality* attribute under *Scheduled Procedure Step Sequence* in MWL is 1, only the *Code Value* from the first item under the *Scheduled Station Class Code Sequence* of a UPS work item is mapped. In case the *Scheduled Station Class Code Sequence* contains more items (and more *Code Value* attribute values underneath) in the UPS workitem, all values except for the first one are omitted from the MWL C-FIND RSP dataset.

Remark (*5):

DICOM Forwarder maps value(s) from the first item only under the *Scheduled Workitem Code Sequence (0040,4018)* sequence of the UPS work item taking into consideration the fact that only single item is supported under the *Scheduled Procedure Step Sequence (0040,0100)* in the MWL worklist.

In case there are multiple items under the *Scheduled Workitem Code Sequence* (e.g. in *Code Value* attributes), all those values (other than the first one) are omitted from the MWL C-FIND RSP dataset.

Remark (*6):

DICOM Forwarder maps value(s) from the first item only under the *Referenced Request Sequence (0040, A370)* of the UPS work item taking into consideration the fact that only single item is supported under the *Requested Procedure Code Sequence (0032,1064)* in the MWL worklist.

In case the *Referenced Request Sequence* contains more items (e.g. under *Code Value* attribute values), all but the first values are omitted from the MWL C-FIND RSP dataset.

Remark (*7):

Requested Procedure ID (0040,1001) from *Referenced Request Sequence (0040,A370)* of UPS workitem is mapped to both *Scheduled Procedure Step ID (0040,0009)* and *Requested Procedure ID (0040,1001)* attribute of the related MWL worklist.

The DICOM Forwarder AE shall return the following status codes in response of a C-FIND Request:

Table 4-38 MWL command return Status Codes

Service Status	Further Meaning	Status Code	HDP Status Code	Behavior
Success		0000	200	The operation is successful.
Success	No content	0000	204	The search completed successfully but returned no results.
Failure	Bad request	C000	400	HDP Core reported that the request was badly formatted.
Failure	Refused: Not authorized	0124	401	Authentication token is expired, or subscription key invalid.
Failure	Forbidden	0112	403	HDP Core reported authorization error.
Failure	Unable to process	C000	404	The specified resource could not be found.
Failure	Unable to process	C000	406	The specified Accept header is not supported.
Failure	Refused: Not authorized	0124	407	Missing or invalid credentials are configured for http proxy server hence ZEISS ID token and/or HDP endpoints are not accessible.
Failure	Resource limitation	A700	429	Too many requests are sent to the HDP.
Failure	Internal server error	0110	500	HDP Core reported that an unexpected error.
Failure	Resource limitation	A702	503	The service is unavailable or busy.

Limitation of C-FIND MWL Queries with large number of matches:

In order to avoid potential timeout issues for the established DICOM DIMSE associations for MWL C-FIND operations, the maximum number of results is limited to 200 workitems.

This number is set by taking into account the maximum allowed size for an http response by the DICOMweb server hosted by HDP.

No errors are reported by DICOM Forwarder in case the number of matching workitems exceeded this value, but only the first 200 results are returned.

4.3 Network Interfaces

4.3.1 Physical Network Interface

The physical network interface is not visible for the instrument application. The instrument application uses the communication stack as offered by the Connectivity Module.

4.3.2 Additional Protocols

Both IP addresses and host names are supported and get resolved.
Else no additional protocols are supported.

4.3.3 IPv4 and IPv6 Support

DICOM Forwarder, on top of Connectivity Modul is released only for IPv4 communication.

4.4 Configuration

Connected device list is configurable in the Web UI of Connectivity Module. Remote AE devices that are not set up and enabled are not allowed to communicate with DICOM Forwarder.

The built in DICOM configuration is:

- AET: CZMADF
- Port number: 11111
- TLS port number: 2762

4.4.1 AE Title/Presentation Address Mapping

The mapping of the AE Title, TCP/IP addresses and ports is configurable and those are stored in the Web UI of Connectivity Module. When a DICOM device is added to this mapping, the corresponding AE title will be allowed to communicate with the DICOM Forwarder AE. Modification or deletion of these devices are also handled in the Web UI of Connectivity Module.

4.4.1.1 Local AE Titles

The IP is not configurable by the DICOM Forwarder, nor the HDP. The IP is administrated by the Connectivity Module. The DICOM Forwarder application uses only one AE for all transactions. The default AE title and port number are given in section 4.4.

4.4.1.2 Remote AE Titles

The mapping of remote AE Titles to TCP/IP addresses and ports is configurable. DICOM Forwarder supports a proprietary auto configuration for corresponding external DICOM devices.

4.4.2 Parameters

4.4.2.1 General Parameters

The general parameters are applied for associations to any of the configured AE Titles from the connected device list.

Table 4-39 Configuration parameters table

Parameter	Default Value	Comment
DICOM Connection Parameters		
backlog	50	Maximum length of the queue of incoming connections
connectTimeout	0	***UNUSED***
requestTimeout	0	Timeout in ms for receiving A-ASSOCIATE-RQ (0 = no timeout)
acceptTimeout	0	Timeout in ms after sending an A-ASSOCIATE-RQ (0 = no timeout)
releaseTimeout	0	Timeout in ms for receiving A-RELEASE-RP (0 = no timeout)
sendTimeout	0	Timeout in ms for sending other DIMSE RQs than C-STORE RQs (0 = no timeout)
storeTimeout	0	Timeout in ms for sending C-STORE RQs. (0 = no timeout)

socketCloseDelay	50	Delay in ms for Socket close after sending A-ASSOCIATE-RJ or A-ABORT
connection.sendBufferSize	0	Set the SO_SNDBUF socket option to specified value in KB.
connection.receiveBufferSize	0	Set the SO_RCVBUF socket option to specified value in KB.
connection.tcpNoDelay	true	Set whether or not this network connection should use Nagle's algorithm as part of its network communication. Values true or false. Boolean "true" if TCP no delay (disable Nagle's algorithm) should be used.
idleTimeout (for inbound connections)	0	Timeout in ms for aborting of idle inbound Associations. (0 = no timeout)
idleTimeout (for outbound connections)	300000	Timeout in ms for aborting of idle outbound Associations. (0 = no timeout)
responseTimeout (for inbound connections)	0	Timeout in ms for receiving other outstanding DIMSE RSPs than C-MOVE or C-GET RSPs inbound connections (0 = no timeout)
responseTimeout (for outbound connections)	300000	Timeout in ms for receiving other outstanding DIMSE RSPs than C-MOVE or C-GET RSPs outbound connections (0 = no timeout)
retrieveTimeout	0	Timeout in ms for receiving outstanding C-MOVE or C-GET RSPs. (0 = no timeout)
retrieveTimeoutTotal	false	Indicates if the timer with the specified timeout for outstanding C-GET and C-MOVE RSPs shall be restarted on receive of pending RSPs.
receivePDULength (for inbound connections)	16378	Set the maximum length (in bytes) of P-DATA-TF PDUs that can be received for inbound connections.
receivePDULength (for outbound connections)	16384	Set the maximum length (in bytes) of P-DATA-TF PDUs that can be received for outbound connections.

sendPDULength (for inbound connections)	16378	Set the maximum length (in bytes) of P-DATA-TF PDUs that will be sent for inbound connections.
sendPDULength (for outbound connections)	16384	Set the maximum length (in bytes) of P-DATA-TF PDUs that will be sent for outbound connections.
maxOpsInvoked	1 (SYNCHRONOUS_MODE)	Set maximum number of outstanding operations this network AE may invoke asynchronously as an SCU.
maxOpsPerformed	1 (SYNCHRONOUS_MODE)	Set the maximum number of operations this network AE may perform asynchronously as an SCP.
packPDV	true	Set whether to send only one PDV in one P-Data-TF-PDU. Values true or false.
Patient Root Q/R and Study Root Q/R SCP Parameters		
Maximum Query Responses (Patient Root Q/R IM and Study Root Q/R IM)	200	
Lenient retrieve mode	True	Accept C-MOVE-RQ without unique keys for all levels above the retrieve level even if support for relational retrieve was not negotiated.

5 *Media Interchange*

Media Interchange is not scope of this document since Media Interchange is not supported by DICOM Forwarder.

6 Support of Character Sets

Accepted Character Sets

Table 6-1 Supported Character Set

Supported Specific Character Set	
Character Set Description	Defined Term
US-ASCII	ISO 2022 IR 6
ISO-8859-1	ISO_IR 100
ISO-8859-2	ISO_IR 101
ISO-8859-3	ISO_IR 109
ISO-8859-4	ISO_IR 110
ISO-8859-5	ISO_IR 144
ISO-8859-6	ISO_IR 127
ISO-8859-7	ISO_IR 126
ISO-8859-8	ISO_IR 138
ISO-8859-9	ISO_IR 148
JIS_X0201	ISO_IR 13
TIS-620	ISO_IR 166
ISO-8859-1	ISO 2022 IR 100
ISO-8859-2	ISO 2022 IR 101
ISO-8859-3	ISO 2022 IR 109
ISO-8859-4	ISO 2022 IR 110
ISO-8859-5	ISO 2022 IR 144
ISO-8859-6	ISO 2022 IR 127
ISO-8859-7	ISO 2022 IR 126
ISO-8859-8	ISO 2022 IR 138
ISO-8859-9	ISO 2022 IR 148
JIS_X0201	ISO 2022 IR 13
TIS-620	ISO 2022 IR 166
JIS0208	ISO 2022 IR 87
JIS0212	ISO 2022 IR 159
cp949	ISO 2022 IR 149
UTF-8	ISO_IR 192
GB18030	GB18030

- By default, DICOMweb server supports ISO_IR 192 character set.
- The C-FIND-RSP dataset provided by DICOM Forwarder contains the specific character set that was stored within matching DICOM file (composite instance) or what was stored within the matching UPS workitem.
- If the client requests a specific character set within C-FIND-RQ and no character set was stored in the matching DICOM composite instance or UPS workitem, then the specific character set of the request is carried over to the C-FIND-RSP. The response is then encoded in the same character set as the request. Characters which are not covered by this character set are substituted with a question mark '?' in the response.
Example: If a client submits a C-FIND request for Patient Name with character set ISO_IR 100, and the name in the stored DICOM dataset is saved as 'Müller', then it would be returned as 'M?ller'.

- If the client did not include a specific character set within C-FIND-RQ and no character set was stored in the matching DICOM composite instance or UPS workitem, then the ISO_IR 192 (UTF-8) specific character set is included in the C-FIND-RSP.

Example: If a client submits a C-FIND request for Patient Name without including specific character set in the C-FIND-RQ and the name in the stored DICOM dataset is saved as 'Müller', then it would be returned as 'Müller'. However, characters which are not covered by the UTF-8 character set will be substituted with a question mark '?'.

Returned Character Sets

When retrieved, DICOM files are returned as they are, no transcoding is done by DICOM Forwarder.

7 Security

7.1 Security Profiles

7.1.1 Security Transport Connection Profiles

The DICOM capabilities of DICOM Forwarder supports BCP 195 RFC 8996 TLS Secure Transport Connection Profile.

The private key and the certificate used by DICOM Forwarder to identify itself in the TLS negotiation with remote application along with its password must be provided in a local keystore file in PKCS12 format on the application host by the Connectivity Module using its Web UI.

The client devices have to be authenticated via a certificate. Public certificates for trusted devices as well as certificates of Certificate Authorities (CA) to validate certificates received from remote applications during the TLS negotiation must be provided in X.509 or PKCS7 format also via the use of the Connectivity Module and its Web UI.

Table 7-1 Transport protocols

Transport protocol	Supported
TLS 1.2	Y
TLS 1.3	Y

Table 7-2 Default Secure Transport Connection Profile

Profile	Creator/Sender	Consumer/Receiver
BCP 195 RFC 8996 TLS Secure Transport Connection Profile	Y	Y

Offered cipher suites in order of preference:

Table 7-3 Supported Cipher Suites

Profile	Cipher Suites	Transport protocol
BCP 195 RFC 8996 TLS Secure Transport Connection Profile	TLS_AES_128_GCM_SHA256	TLS 1.3
	TLS_AES_256_GCM_SHA384	TLS 1.3
	TLS_CHACHA20_POLY1305_SHA256	TLS 1.3
	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	TLS 1.2
	TLS_DHE_RSA_WITH_AES_256_GCM_SHA384	TLS 1.2
	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	TLS 1.2
	TLS_DHE_RSA_WITH_AES_128_GCM_SHA256	TLS 1.2

7.1.2 Basic Application Level Confidentiality Profile

Not supported.

7.1.3 Association Level Security

DICOM Forwarder accepts only associations from configured remote AE Titles.

7.1.4 Application Level Security

The DICOM capabilities of DICOM Forwarder allows the use of either a conventional (non-secure) DICOM communication or a secure DICOM communication based on the Transport Layer Security (TLS) protocol. Additionally,

- It is assumed that DICOM Forwarder is used within a secured environment. It is assumed that a secured environment includes at a minimum Firewall or router protections to ensure that only approved external hosts have network access to DICOM Forwarder
- Firewall or router protections to ensure that DICOM Forwarder only has network access to approve external hosts and services.
- Any communication with external hosts and services outside the locally secured environment use appropriate secure network channels (e.g. such as a Virtual Private Network (VPN))
- Other network security procedures such as automated intrusion detection may be appropriate in some environments. Additional security features may be established by the local security policy and are beyond the scope of this conformance statement.

Apart from these, the DICOM capabilities of the DICOM Forwarder does not support any specific security measures. Although the DICOM Forwarder is part of the Connectivity Module provided and HDP ecosystem, the application itself is restricted to run in a separate containerized environment.

8 Annexes

8.1 IOD Contents

8.1.1 Created SOP Instance(s)

DICOM Forwarder does not create any DICOM data.

8.1.2 Usage of Attributes from Received IOD's

Not applicable.

8.1.3 Attribute Mapping

DICOM Forwarder translates DICOM DIMSE MWL Request to DICOM Web UPS-RS Request. The attribute mapping is applied on the **Table 4-32**.

8.1.4 Coerced/Modified Fields

DICOM Forwarder does not modify any DICOM data.

8.2 Data Dictionary of Private Attributes

DICOM Forwarder does not create any private attributes.

8.3 Coded Terminology and Templates

DICOM Forwarder does not use or create any codes.

8.4 Greyscale Image Consistency

Not applicable.

8.5 Standard Extended / Specialized/ Private SOP Classes

Not applicable

8.6 Private Transfer Syntaxes

No Private Transfer Syntax is supported.



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